

Praveen Kumar

Technical Architect | Leeds, United Kingdom

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PROFESSIONAL SUMMARY

Technical Architect with hands-on experience as an Azure Solutions Architect & Azure AI Engineer Associate, delivering enterprise-scale cloud and AI platforms across the UK and India. Specialist in Microsoft Azure, Foundry, Agentic AI architectures, RAG systems, and LLM evaluation frameworks, with hands-on depth in Microsoft Agent Framework (MAF), multimodal agents, and MCP server implementations, several shared as [open-source](#) reference projects. Architected and delivered complex Azure and AI solutions for FTSE 100 clients including [ASDA](#), [BP](#), [AON](#), [DLL](#), [CAPITA](#), and [Airbus](#), driving faster time-to-production for cloud and AI platforms, underpinned by delivery of high-quality web and API products at scale (27M+ monthly users) using Azure PaaS, Infrastructure as Code (Terraform), and observability tooling (Azure Monitor/Application Insights). Passionate about clean, testable code, TDD, security best practices, and cross-functional collaboration in Agile teams to deliver user-focused solutions that drive business value.

CORE SKILLS

- **Agentic AI & Foundry:** Microsoft Foundry, Foundry Agent Service, Agentic Workflows, Multi-Agent Pipelines, LangChain, LangGraph, LangSmith, AutoGen, CrewAI, Semantic Kernel, MCP, Microsoft Agent Framework (MAF).
- **Generative AI & LLMs:** RAG, Prompt Engineering, Fine-Tuning, Vector/Semantic Search, Embeddings, LLM Evaluation, Red Teaming, Guardrails, Azure AI Content Safety, Agentic AI, Model Context Protocol (MCP), Retrieval-Augmented Generation Pipelines, LLM Orchestration, Chunking Strategies, Hybrid Search, Model Benchmarking, Hallucination Mitigation, Token Optimization, Multimodal AI (Vision + Language), Function/Tool Calling.
- **Azure AI Services:** AI Search, Document Intelligence, Vision, Speech, Language, Translator, Cognitive Services, Hugging Face.
- **Azure Cloud & Architecture:** Azure Functions, API Management, App Services, Cosmos DB, Event Hub, Service Bus, ACR, ACL, Azure Firewall, Application Gateway, VNET/NSG, Key Vault, Storage, Microsoft Entra ID, Azure Kubernetes Service (AKS), Azure SQL Database, Azure Monitor, Application Insights, Log Analytics, Logic Apps, Event Grid, Azure Load Balancer, Traffic Manager, Private Link/Endpoints, Managed Identities, Azure Policy, Cost Management, Well-Architected Framework.
- **Engineering & DevOps:** Python, C#, .NET Core, T-SQL, JavaScript, Azure DevOps, Terraform, Jenkins, CI/CD, Containerization, Observability.
- **Architecture & Delivery:** Microservices, Serverless, Event-Driven, Enterprise Architecture, Agile, Pre-Sales, Stakeholder Management.
- **Web Application Development:** APIs, JavaScript, HTML, CSS, jQuery.

PROFESSIONAL EXPERIENCE

Technical Architect — [TCS](#), United Kingdom

Apr 2021 – Present

- Architected a scalable Azure PaaS platform for ASDA, supporting the separation of 2,500+ systems from Walmart and improving e-commerce throughput by 30% through event-driven microservices and Azure API Management (APIM), exposing high-traffic applications for public and private access with 99% uptime and a 35% efficiency gain, driving a 28% boost in user engagement and a 42% reduction in page load times.
- Designed and delivered RAG-based conversational AI systems using Microsoft Foundry and Azure AI Search and built multi-agent AI pipelines with LangGraph and Foundry Agent Service, including supervisor routing, retry logic, and LangSmith observability.
- Developed and published open-source Microsoft Foundry reference repositories (30+ evaluation scripts, multimodal agents, function calling, agent memory, MCP integration), widely used as reference material by developers building enterprise AI agents on Azure.
- Delivered Microsoft Purview-integrated metadata ingestion pipelines for BP, enabling automated data lineage and governance across distributed data estates.
- Led architecture workshops, technical reviews, and Agile delivery teams across multiple concurrent client engagements; mentored teams on cloud automation practices, reducing cloud spend by 12%, and led incident response efforts using automated runbooks and root cause analysis (RCA), recognized with a TCS Top Performer Award for mentorship and engineering innovation.

Technical Lead — [Wipro Ltd.](#), Pune, India

Jul 2020 – Apr 2021

- Directed cloud modernization for UnitedHealth Group, migrating legacy systems to Azure PaaS using microservices and Azure Functions, reducing infrastructure cost and improving API performance.
- Designed secure APIs using .NET Core and Azure API Management with OAuth 2.0 and RBAC. Automated CI/CD pipelines using Azure DevOps and Terraform, improving deployment cycle time and release reliability.

Consultant — [Atos India Pvt. Ltd.](#), India

Aug 2013 – Mar 2020

- Delivered digital transformation programmes for WSP, DLL, and Airbus using Azure PaaS and .NET Core.
- Built multilingual customer engagement platforms on Sitecore CMS, supporting 10+ languages across client markets.
- Managed Agile teams of 8–12 engineers across design, development, and testing; contributed to pre-sales, solution proposals, and client technical presentations.
- Contributed to pre-sales activities and solution proposals, supporting business development efforts.
- Delivered client technical presentations, communicating solution architecture and value to stakeholders.

Senior Software Engineer — [Sopra Group](#), India

Jun 2011 – Jul 2013

- Delivered Azure-based digital platforms for CAPITA and Airbus, including portals, dashboards, and enterprise integrations, using services such as Azure App Service and Azure Functions.
- Contributed to solution architecture and development, building API-based integrations to connect enterprise systems and meet client requirements.
- Managed development and testing workflows using Azure DevOps within Agile teams, ensuring quality and timely delivery of each release.

Software Engineer — OnDemand Inc. / ODO Software Solutions Pvt. Ltd., India

May 2008 – Jun 2011

- Contributed to the architecture and design of ASP.NET applications, collaborating with Agile teams to define technical solutions.
- Developed and implemented application features, ensuring alignment with project requirements and coding standards.
- Conducted testing (unit/integration) to ensure application quality, reliability, and performance prior to release.

EDUCATION

Master of Computer Applications (MCA) — [United Institute of Management](#), 2008

Bachelor's Degree — [Sachidanand Sinha College](#), 2003

CERTIFICATIONS

- [Azure Solutions Architect Expert \(AZ-305\)](#)
- [Azure Administrator Associate \(AZ-104\)](#)
- [Azure AI Engineer Associate \(AI-102\)](#)

KEY ACHIEVEMENTS

- Delivered 18+ enterprise-grade applications (web, API, batch, console) using Azure PaaS, .NET and Python supporting high-traffic environments (27M+ monthly users), achieving a 99.5% deployment success rate across 160+ Azure resources via IAC and automated CI/CD.
- Reduced critical vulnerabilities by 75% through integrated security scanning (Snyk, Trivy) across hundreds of repositories.
- Created end-to-end infrastructure for 8+ applications covering core Azure services (Microsoft Entra ID, App Services, Azure SQL, APIM, vNets, NSGs, Application Gateway, Key Vault, monitoring, Storage, Service Bus, Event Hub).
- Reduced cloud operational spend by 12% through automation of routine cloud management tasks and mentorship of engineering teams on cost-optimization best practices.
- Improved incident response times by 27% by introducing automated runbooks and standardized root cause analysis (RCA) processes across client engagements.
- Recognized with a TCS Top Performer Award for engineering innovation and technical mentorship across concurrent client programmes.
- Improved RAG response accuracy by 17% through refined chunking, hybrid search, and retrieval evaluation frameworks built on Azure AI Search.
- Reduced LLM inference costs by 19% via prompt optimization, caching strategies, and token-usage monitoring across multi-agent pipelines.